Sustainable Information Systems and Green Metrics

Introduction

It is estimated that information and communication technology (ICT) is responsible for at least 2% of global greenhouse gas (GHG) emissions. The use of IT for greening will play a key role in the delivery of benefits that can alleviate an estimated five times the GHG footprint of IT itself.

Relevant and accurate data, information, metrics and key performance indicators (KPIs) are critical to supporting sustainable practices, and the need to manage this information has led to the emergence of green information systems (GIS) as a field in itself. There is substantial potential for green IS to bring together business processes, resource planning, direct and in-direct activities and extended supply chains to effect positive changes across the entire activities of governments, organizations and individuals. GIS is the engine driving both the strategic and operational management of sustainability.

As sustainable information is needed at both the macro and micro levels, it will require a multilevel approach that provides information and metrics that can drive high-level strategic corporate and regional sustainability plans as well as low-level actions like improving the energy efficiency of an office worker.

Multilevel Sustainable Information

Sustainability requires information on the use, flows and destinies of energy, water and materials including waste, along with monetary information on environment-related costs, earnings and savings. This type of information is critical if we are to understand the causal relationships between the various actions that can be taken and their impact on sustainable performance. However, the problem is broad in scope and the necessary information may not be available, or may be difficult to collect. Improving sustainability performance, especially through changing the way an organization or activity operates, requires a number of practical steps which will include the need for a systematic approach for information gathering and analysis.

In order to tackle the problem, it is important to break it down into smaller, manageable pieces by adding boundaries and scopes to activities to make them more manageable. A 'boundary' or 'scope' can be drawn narrowly for a specific activity (e.g. to consider only the emissions that arise directly from that activity) or broadly (including emissions indirectly associated with the activity).

The problem of sustainability from the perspective of a multilevel information problem, where individual levels are used to provide a scope and boundary for activities is shown in Table below.

Table 9.1 Multilevel sustainable information

Level	Stakeholders	Information requirements	Information regarding
Regional/City	Public administration, policymakers, politicians, corporations, citizens, and regulators	Regional sustainability plans, movement and transport, biodiversity and environmental impacts, health and well being, resource management, waste disposal	Regional energy consumption Energy consumption by sector GHG emissions per capita GHG intensity of energy consumption
Organizational	Executive team, shareholders, citizens, regulators, suppliers and consumers	Corporate sustainable reports, sustainability plans, sustainable business objectives, business function performance, and so on	Total energy consumption Total electricity consumption Percentage of renewable energy sources
Functional	Function management team, organization management and employees	Function sustainability performance, functions sustainability objectives and so on	Facility electricity consumption Internal electricity consumption DC electricity consumption
Individual	Citizens and employees	Direct and indirect cost of actions of the individual, private travel, energy use, work commute and impacts of products and services consumed	Business travel Employee commuting Employee IT energy use
Product or service	Product producers and consumers	Analysis of the sustainability impacts of the product or service, manufacturing cost, usage cost, disposal cost and so on	Server electricity consumption Laptop electricity consumption

Sustainability Hierarchy Models

Sustainability is a complex and broad subject. In order to make sense of the various frameworks and tools, a hierarchy of sustainability models that is illustrated in Figure.

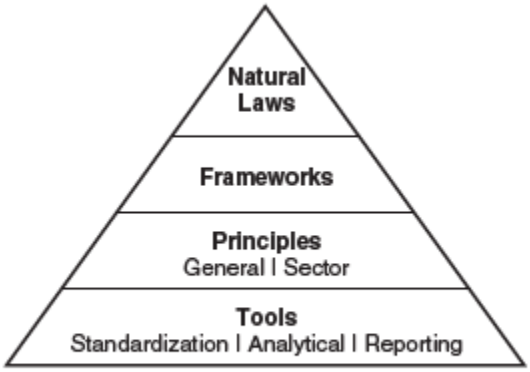


Figure 9.1 Hierarchy of sustainability models (adapted from Hitchcock and Willard, 2008).

- Natural laws: The laws imposed upon us by Mother Nature such as the Law of Conservation of Mass and Energy, Law of Entropy and Laws of Thermodynamics.
- Frameworks: High-level conceptual rules for sustainability that conform to natural laws.
- Principles: General and sector-specific guidelines that detail sustainable practices and actions.
- Tools: Methodologies, standards and strategies for implementation.

Sustainability Frameworks

In order to improve the understanding and communication of sustainability issues and to provide high-level definitions of sustainability, a number of frameworks have been proposed. Natural Capitalism, The Natural Step, Ecological Footprint and the triple bottom line (TBL) are popular frameworks. The frameworks can be used to develop high-level visions and planning for sustainability activities.

Natural Capitalism

The concept of Natural Capitalism, an extension of the economic concept of capital to the natural environment, is to look at the environment as a system that yields a valuable flow of goods and services (e.g. fish, trees and other such things). Natural capitalism requires the following shifts in business practices: Radically increase productivity in the use of natural resources; shift to biologically inspired production models and materials; move to a 'service-and-flow' business model; and reinvest in natural capital.

The Natural Step

The objective of The Natural Step is to help reduce the potential causes of environmental problems. This framework defines four system conditions, derived from the Laws of Thermodynamics, for a sustainable society. In a sustainable society, nature's functions and diversity must not be systematically subjected to (1) increasing concentrations of substances extracted from the Earth's crust, (2) increasing concentrations of substances produced by society, (3) degradation by physical means and (4) resources must be used fairly and efficiently to meet the basic needs of people worldwide.

Table 9.2 The natural step framework: system conditions and sustainability principles

System conditions	Sustainability principles
Concentrations of substances extracted from the Earth's crust	Eliminate our contribution to the progressive build-up of substances extracted from the Earth's crust (e.g. heavy metals and fossil fuels)
Concentrations of substances produced by society	Eliminate our contribution to the progressive build-up of chemicals and compounds produced by society (e.g. dioxins, polychlorinated biphenyls (PCBs) and dichlorodiphenyltrichloroethane (DDT))
Degradation by physical means	Eliminate our contribution to the progressive physical degradation and destruction of nature and natural processes (e.g. over-harvesting forests and paving over critical wildlife habitat)
In that society, people not subject to conditions that systemically undermine their capacity to meet their needs	Eliminate our contribution to conditions that undermine people's capacity to meet their basic human needs (e.g. unsafe working conditions and not enough pay to live on)

Ecological Footprint

The ecological footprint is the measure of environmental impact of particular actions on the Earth's natural resources and ecosystem functionality. It compares human demand with the Earth's ecological capacity to regenerate. The assessment can be used at the individual, city or regional level to estimate how much of the Earth (or how many planet Earths) it would take to support humanity if everybody lived a given lifestyle.

Triple Bottom Line (TBL)

The TBL framework allows an organization to focus on not only its economic bottom line, but also its environmental and social 'bottom lines'. TBL expands the scope of responsibility for an organization from shareholders and owners to also include stakeholders – anyone who is influenced, either directly or indirectly, by the actions of the organization.

Sustainability Principles

Below the high-level frameworks are a broad range of principles that provide guidance on the sustainability practices needed to achieve the objectives set out in higher level frameworks. Principles are typically created via a group or community process with the aim to produce a number of aspirational statements to provide guidance on sustainability. Two well-known general principles, are the 3 Rs and Earth Charter.

Reduce, Reuse and Recycle (3R's)

Reduce, Reuse and Recycle are three waste management strategies that are used collectively within a waste hierarchy known as the 3 Rs. The 3 Rs classify the strategies according to the desired order of use: It is better to reduce than reuse, and better to reuse than recycle. The aim of the waste hierarchy is to extract the maximum practical benefits from products and to generate the minimum amount of waste.

Earth Charter

The Earth Charter is an international declaration on the principles needed to build a just, sustainable and peaceful global society. The Earth Charter proposes that environmental protection, human rights, equitable human development and peace are interdependent and indivisible.

Tools for Sustainability

At the bottom of the hierarchy are the sustainability tools that provide guidance and best practices. Sustainability tools come in many forms, from methodologies used to help determine the environmental impact of a steel water bottle, to certifying the design and construction of a building. Sustainability tools can be general purpose, such as the GHG Protocol for emissions reporting, or specific to a sector, like the Leadership in Energy and Environmental Design (LEED) for green building certification. These tools provide sustainability information needed to make sustainable decisions and drive sustainable behavior. Many of these tools can be classified as IT for green or as an opportunity for IT to 'green' processes or activities outside of IT.

Product Level Information

Understanding the impacts of a product or service requires an analysis of all potential impacts associated with a product, process or service for its entire life cycle. This is achieved using a technique known as life cycle assessment.

Life-Cycle Assessment

Life-cycle assessment (LCA), also known as life cycle analysis, is a technique to systematically identify resource flows and environmental impacts associated with all the stages of product and service provision. LCA provides a quantitative cradle-to-grave analysis of the products or services' global environmental costs (i.e. from raw materials through materials processing, manufacture, distribution, use, repair and maintenance and disposal or recycling). The demand for LCA data and tools has accelerated with the growing global demand to assess and reduce GHG emissions from different manufacturing and service sectors.

LCA can be used as a tool to study the impacts of a single product to determine the stages of its life cycle with most impact. LCA can also be used as decision support when determining the environmental impact of two comparable products or services.

The Four Stages of LCA

For a LCA, both ISO 14040 (ISO, 2006a) and 14044 (ISO, 2006b) standards follow four distinct phases process:

1. Goal and scope definition: It is important to ask the right question to ensure whether the LCA is successful. The first step in this process is the framing of the key questions for the assessment. Typical steps include defining the goal(s) of the study, determining what type of information is needed to inform decision makers, defining functional units (environmental impact, energy efficiency, life span, cost per use, etc.), defining the system boundaries and studying perspective, allocation principles, environmental impact assessment categories and level of detail.
2. Inventory analysis: The second phase involves data collection and modelling of the product and service system with process flow models and inventories of resource use and process emissions. The data must be related to the functional unit defined in the goal and scope definition and include all data related to environmental (e.g. CO_2) and technical (e.g. intermediate chemicals) quantities for all relevant unit processes within the study boundaries that compose the product system. Examples of inputs and outputs include materials, energy, chemicals, air emissions, water emissions, solid waste, radiation and land use. The results of a life cycle inventory provide verified information about all inputs and outputs in the form of elementary flow to and from the environment from all the unit processes involved in the study.
3. Impact assessment: The third phase evaluates the contribution to selected impact assessment categories, such as 'climate change', 'energy usage' and 'resource depletion'. Impact potential of the inventory is calculated and characterized according to the categories. Results can then be normalized across categories (same unit) and weighted according to the relative importance of the category; both of these actions are voluntary according to the ISO standard.
4. Interpretation: The final phase involves interpretation of the results to determine the level of confidence and communicate them in a fair, complete and accurate manner. This is accomplished by identifying the data elements that contribute significantly to each impact category, evaluating the sensitivity of these significant data elements,

assessing the completeness and consistency of the study and drawing conclusions and recommendations based on a clear understanding of how the LCA was conducted and the results were developed.

CRT Monitors versus LCD Monitors: Life Cycle Assessment

A comprehensive environmental LCA of the traditional cathode ray tube (CRT) and newer liquid crystal display (LCD) monitors was conducted through the EPA's Design for the Environment (DfE) programme. The objective of the study was to evaluate the environmental and human health life cycle impacts of functionally equivalent 17 in. CRT and 15 in. LCD monitors. The study assessed the energy consumption, resources input and pollution produced over the lifetime of the equipment. The cradle-to-grave analysis was divided into three stages: (i) cradle to gate (manufacturing), (ii) use and (iii) end of life (disposing or reusing). Each stage was assessed for the energy consumed, materials used in manufacturing and associated waste. Components manufactured in different locations, where energy sources can differ due to the way local energy is produced, such as coal versus nuclear, were taken into account.

LCA concluded that LCD monitors are about 10 times better for resource usage and energy use, and five times better for landfill use. However, LCDs are only 15% better for global warming because the LCD manufacturing process uses Sulphur hexafluoride, a significant GHG.

Individual Level Information

Primary sources of emissions for an individual will include the life cycle of products and services they purchase and use, the construction (LCA of material and construction) and operation of their residence (energy use, water use and waste disposal and recycling) and private travel (especially emissions from long-haul flights).

The ability for individuals to understand the impacts of their activities is critical. Online carbon calculators are a popular mechanism to calculate impact; however, these may have significant limitation including static calculation, a lack of personalization and limited capacity for historical analysis. However, tools are improving for individuals. For instance, home energy usage is an area that has seen the early adoption of real-time monitoring systems with the on-going development of smart meters and the Smart Grids. Studies have shown that energy conservation improves if residents receive real-time feedback on their energy usage.

A significant contribution to CO_2 emissions levels is the daily commute by an individual. In response to addressing this challenge, a group of Volvo employees developed an IT solution that would motivate people to reduce the emission cost of their daily commute by measuring the time, efficiency and environmental impact of their commuting, by using a small application on mobile phones. This app was transformed into a global service and launched as Commute Greener!, which is able to help companies, cities and individuals around the world understand how to commute in a greener way.

Efforts such as Commute Greener and Smart Grids show how tracking information at the individual level can benefit the individual directly, and also be aggregated at a higher level of analysis to benefit a region or community.

Functional Level Information

An organization typically comprises multiple functions or operating divisions and departments including marketing, finance, operations, human resources and research and development.

Sustainability issues at the functional level are related to business processes and the value chain; they involve the development and coordination of resources through which the overall organization's objectives and goals can be executed efficiently and effectively. A close relationship exists between organization level and functional level metrics, and it is important that metrics at both levels be aligned.

Monitoring energy usage for an IT function can involve metrics around building energy usage (lights and heat), staff computer energy usage and in particular energy used within data centers.

Data Centre Energy Efficiency

The power needs of data centers may range from a few kilowatts for a rack of servers in a closet, to several tens of megawatts for large facilities. Power usage in a data center goes beyond the direct power needs of servers to include networking, cooling, lighting and facilities management. The US Environmental Protection Agency (EPA)

estimates that servers and data centers are responsible for up to 1.5% of the total US electricity consumption. With electricity costs the dominant operating expense of a data center, it is vital to maximize the center's operational efficiency to reduce both the environmental and economic costs.

Data Centre Power Metrics

Developed by the Green Grid, power usage effectiveness (PUE) is a measure of how efficiently a data center uses its power. PUE measures how much power the computing equipment consumes in contrast to cooling and other overheads uses. PUE is defined as follows:

$$PUE = Total\ Facility\ Power/IT\ Equipment\ Power$$

The reciprocal of PUE is data center infrastructure efficiency (DCiE) and is defined as follows:

$$DCiE = 1/PUE = IT\ Equipment\ Power/Total\ Facility\ Power \times 100\%$$

IT equipment power includes the load associated with all of the IT equipment, such as compute, storage and network equipment, along with supplemental equipment used to monitor or otherwise control the data center including Keyboard Video Mouse (KVM) switches, monitors, workstations and laptops. Total facility power includes everything that supports the IT equipment load such as power delivery (uninterruptible power supply (UPS), generators, batteries, etc.), a cooling system (chillers, computer room air conditioning units (CRACs), direct expansion air handler (DX) units, pumps and cooling towers), compute, network, storage nodes and other loads such as data center lighting.

Emerging Data Centre Metrics

A number of metrics are under development to measure dimensions including resource utilization and environmental impact. Each of these metrics provides data center operators more visibility into where opportunities for further efficiency improvements exist.

• Carbon usage effectiveness (CUE): Measures data center–specific carbon emissions, which are emerging as an extremely important factor in the design, location and operation of a data center. CUE, combined with PUE, can assess the relative sustainability of a data center to determine if any energy efficiency and/or sustainable energy improvements need to be made.

$$CUE = CO_2\ emitted\ (\mathrm{kgCO_2\ eq})/unit\ of\ energy\ (\mathrm{kWh})) \times (Total\ Data\ Centre\ Energy/IT\ Equipment\ Energy)$$

• Energy reuse effectiveness (ERE): Many data centers are now recovering waste energy from their operations and reusing it outside of the data center such as heating office space or homes. Since PUE does not consider these alternate uses for waste energy, the ERE metric is used.

$$ERE = (Total\ Facility\ Power - Power\ Reuse)/IT\ Equipment\ Power$$

• Data center energy productivity (DCeP): DCeP quantifies useful work compared to the energy it requires. It can be calculated for an individual IT device or a cluster of computing equipment. DCeP is a sophisticated metric where useful work and energy are compared relative to a user-defined time limit.

$$DCeP = Useful\ Work\ Produced/Total\ Data\ Centre\ Energy\ Consumed\ over\ time$$

• Data center computer efficiency (DCcE): DCcE – and its underlying sub-metric, server compute efficiency (ScE), enables data center operators to determine the efficiency of their compute resources, which allows them to identify areas of inefficiency. The metric can reveal unused compute resource within a data center making it easier for data center operators to discover unused servers (both physical and virtual) and then decommission or redeploy them.

• Environmental consumer chargeback: Carbon dioxide and environmental damage are gaining acceptance as viable chargeable commodities. Using environmental chargeback models data center operators can "chargeback"

the environmental impacts (i.e. CO_2 emissions), in addition to the financial costs, of their services to the consuming end-users. Environmental chargebacks can have a positive effect on environmental impacts by linking consumers to the indirect impacts of their service usage, allowing them to understand the impact of their actions.

Organizational Level Information

Sustainability strategy and direction are determined at the organization level based on macro concerns; identifying the overall sustainability goals of the organization, determining the types of businesses and activities in which the organization should be involved and defining organizational responsibilities.

Organizations need to determine their goals and objectives for sustainability. Typically, organizational sustainability goals involve one or more of the following:

- Develop significant capabilities and a reputation for sustainability leadership.
- Keep pace with industry or stakeholder expectations.
- Meet minimum compliance requirements and reap readily available benefits.

Agreeing on one's desired business posture on sustainability will have a significant impact on business and thus on necessary goals and priorities. It is important to be clear about the organization's business objectives and the role of sustainability in enabling those objectives. Performance metrics and KPIs are used to measure environmental, social and economic impacts, and to ensure the delivery of strategic (and sustainability-related) objectives.

A sample of widely used TBL KPIs typically found in organization level sustainability reports, also known as integrated reporting, and is provided in Table below.

Table 9.4 A summary of widely used KPIs in sustainability reports

Economic indicators	Environmental indicators	Social indicators
Revenues	Investments in environmental protection	Raises awareness and upholds organization's values, code of conduct and principles
Operating costs	Greenhouse gas emissions	Total workforce by employment type, contract and region
Operating margin	Spills	Hours worked
Operating profit	Water use	Employee turnover
Employee wages	Waste (hazardous and nonhazardous)	Minorities in management
Employee benefits	Employee commuting	Benefits for full-time and part-time employees
Payments to providers of capital	Environmental and polluting fines	Employee education and training expenditure
Payments to government in taxes	Total energy consumption	Percentage of employees receiving regular performance and career reviews
Research and development	Renewable energy consumption	Flexible working arrangements
Sustainable innovations	Facility energy consumption	Employee satisfaction
Customer satisfaction	Internal energy consumption	Remuneration and bonuses related to sustainable development
Product safety and quality	Data centre energy consumption	Injuries and accidents
Capital and exploration expenditure	Business travel (flights)	Child labour
Sales and marketing	Fleet fuel consumption	Community contribution
General and administration	Paper consumption	Losses of customer data
	Noncompliance with regulations and voluntary codes	
	Sites with a (certified) environmental management system	

Reporting Greenhouse Gas Emissions

The GHG Protocol is the most widely used international accounting tool for government and business leaders to understand, quantify and manage the emissions of all six major GHGs: carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs) and sulphur hexafluoride (SF_6). Many of

these gases have a 'global warming potential' that is many times greater than that of CO_2. The GHG emissions generated directly and indirectly by an organization can be classified into three 'scopes', based on the source of the emissions. The scope of emissions is illustrated in Figure below.

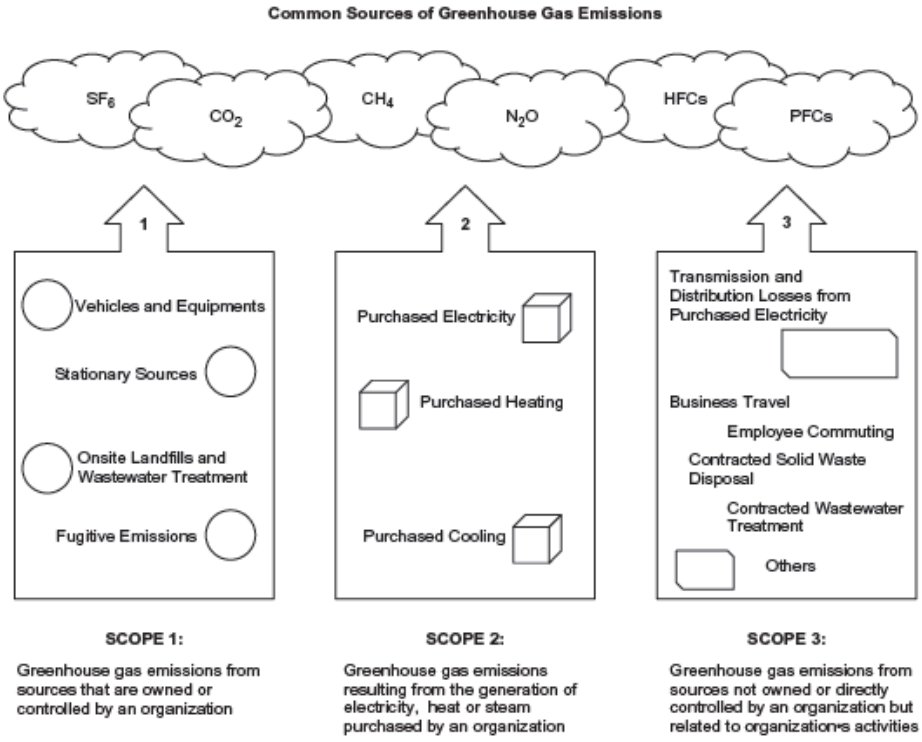


Figure 9.2 Scopes of greenhouse gas emissions.

The GHG Protocol is based on five principles (relevance, completeness, consistency, transparency and accuracy) and lays out a sequence of steps that organizations should follow to account for and report their emissions of GHG. These include the following:

1. Defining the geographical boundaries within which the organization operates.
2. Setting organizational boundaries.
3. Identifying reporting entities at the corporate and facility levels.
4. Deciding emission scopes (1, 2 and/or 3) and which GHGs are to be reported.
5. As appropriate, adopting estimating protocols specific to their sector(s) of industry.
6. Establishing a base year and setting targets for future years.

Regional/City Level Information

Understanding the relationship between economic, environmental, and social factors is essential for the authorities within a region to manage their approach to sustainability.

Strong leadership and governance plays a key role in enhancing a regions sustainability through plans and policies that enhancing resource efficiency and embrace new technologies to improve levels of sustainability through careful management and planning. To ensure that the region is moving towards sustainability the baseline and trend information across all aspects of the region must be made using indicators to track the progress of key aspects of the region such as energy use, total employment rate or level of mental wellbeing, and to use this information to effect positive change. These indicators should be contextualized with a set of region statistics against which the indicators can be measured or analyzed (i.e. population, population density, age dependency ratios, geographical size, national and regional gross domestic product and gross national product, consumer price indices, life expectancy, etc.).

The Carbon Disclosure Project has a wealth of experience in monitoring the GHG emissions from companies; they are extending the project to the city scale. Carbon Disclosure Project for Cities initiative allows cities to disclose their emissions data and benchmark their performance against other cities.

Dublin City Council operationalized the framework through an implementation schema composed of five specific levels: systems level, purpose level, strategic level, actions level and tools level.

- The City Council developed a governance model based on these levels: Joined-up systems, Vision, Strategy, Actions and Tools.
- The integrated approach is useful as it helps avoid the tendency in planning to focus only on a subset of issues or areas ignoring broader, connected issues leading to a need to expand the system boundaries'.
- Examples of indicators used to support the Dublin City Council sustainability plan are based on Transport, waste recycling, water use, biodiversity & parks, energy use.

Measuring the Maturity of Sustainable ICT

A Capability Maturity Framework for SICT

The SICT–CMF gives organizations a vital tool to manage their sustainability capability. The framework provides a comprehensive value-based model for organizing, evaluating, planning and managing SICT capabilities. Using the framework, organizations can assess the maturity of their SICT capability and systematically improve capabilities in a measurable way to meet their sustainability objectives.

The SICT–CMF assessment methodology determines how SICT capabilities are contributing to the business organization's overall sustainability goals and objectives. This gap analysis between what the business wants and what SICT is actually achieving positions the SICT–CMF as a management tool for aligning SICT capabilities with business sustainability objectives.

The framework focusses on the execution of four key actions for increasing SICT's business value:

- Define the scope and goal of SICT.
- Understand the current SICT capability maturity level.
- Systematically develop and manage the SICT capability building blocks.
- Assess and manage SICT progress over time.

Defining the Scope and Goal

Firstly, the organization must define the scope of its SICT effort. As a prerequisite, the organization should identify how it views sustainability and its own aspirations. Typically, organizational goals involve one or more of the following:

- Develop significant capabilities and a reputation for environmental leadership.
- Keep pace with industry or stakeholder expectations.
- Meet minimum compliance requirements and reap readily available benefits.

Secondly, the organization must define the goals of its SICT effort. It is important to be clear on the organization's business objectives and the role of SICT in enabling those objectives. Having a transparent agreement between business and IT stakeholders can tangibly help achieve those objectives. Significant benefits can be gained even by simply understanding the relationship between business and SICT goals.

Capability Maturity Levels

The framework defines a five-level maturity curve for identifying and developing SICT capabilities:

1. Initial: SICT is ad hoc; there is little understanding of the subject and few or no related policies. Accountabilities for SICT are not defined, and SICT is not considered in the system's life cycle.

2. Basic: There is a limited SICT strategy with associated execution plans. It is largely reactive and lacks consistency. There is an increasing awareness of the subject, but accountability is not clearly established. Some policies might exist but are adopted inconsistently.

3. Intermediate: A SICT strategy exists with associated plans and priorities. The organization has developed capabilities and skills and encourages individuals to contribute to sustainability programmes. The organization includes SICT across the full system's life cycle, and it tracks targets and metrics on an individual project basis.

4. Advanced: Sustainability is a core component of the IT and business-planning life cycles. IT and business jointly drive programmes and progress. The organization recognizes SICT as a significant contributor to its sustainability strategy. It aligns business and SICT metrics to achieve success across the enterprise. It also designs policies to enable the achievement of best practices.

5. Optimizing: The organization employs SICT practices across the extended enterprise to include customers, suppliers and partners. The industry recognizes the organization as a sustainability leader and uses its SICT practices to drive industry standards. The organization recognizes SICT as a key factor in driving sustainability as a competitive differentiator.

This maturity curve serves two important purposes. Firstly, it is the basis of an assessment process that helps to determine the current maturity level. Secondly, it provides a view of the growth path by identifying the next set of capabilities an organization should develop to drive greater business value from SICT.

SICT Capability Building Blocks

Whilst it is useful to understand the broad path to increasing maturity, it is more important to assess an organization's specific capabilities related to SICT. The SICT framework consists of nine capability building blocks (see Table) across the following four categories:

- Strategy and planning, which includes the specific objectives of SICT and its alignment with the organization's overall sustainability strategy, objectives and goals.
- Process management, which includes the sourcing, operation and disposal of ICT systems, as well as the provision of systems based on sustainability objectives and the reporting of performance.
- People and culture, which defines a common language to improve communication throughout the enterprise and establishes activities to help embed sustainability principles across IT and the wider enterprise.
- Governance, which develops common and consistent policies and requires accountability and compliance with relevant regulation and legislation.

Table 9.8 Capability building blocks of SICT

Category	Capability building block	Description
Strategy and planning	Alignment	Define and execute the ICT sustainability strategy to influence and align to business sustainability objectives.
	Objectives	Define and agree on sustainability objectives for ICT.
Process management	Operations and life cycle	Source (purchase), operate and dispose of ICT systems to deliver sustainability objectives.
	ICT-enabled business processes	Create provisions for ICT systems that enable improved sustainability outcomes across the extended enterprise.
	Performance and reporting	Report and demonstrate progress against ICT-specific and ICT-enabled sustainability objectives, within the ICT business and across the extended enterprise.
People and culture	Adoption	Embed sustainability principles across ICT and the extended enterprise.
	Language	Define, communicate and use common sustainability language and vocabulary across ICT and other business units, including the extended enterprise, to leverage a common understanding.
Governance	External compliance	Evangelize sustainability successes and contribute to industry best practices.
	Corporate policies	Enable and demonstrate compliance with ICT and business sustainability legislation and regulation. Require accountability for sustainability roles and decision making across ICT and the enterprise.

The first step to systematically develop and manage the nine capabilities within this framework is to assess the organization's status in relation to each one. The assessment begins with a survey of IT and business leaders to understand their individual assessments of the maturity and importance of these capabilities.

The assessment also provides valuable insight into the similarities and differences in how key stakeholders view both the importance and maturity of individual capabilities as well as the overall vision for success.

Plotting current levels of maturity and strategic importance lets an organization quickly identify gaps in capabilities. This is the foundation for developing a meaningful action plan. Figure shows the results of an organization's assessment of the importance of capabilities versus its own assessment of its current maturity in those capabilities.

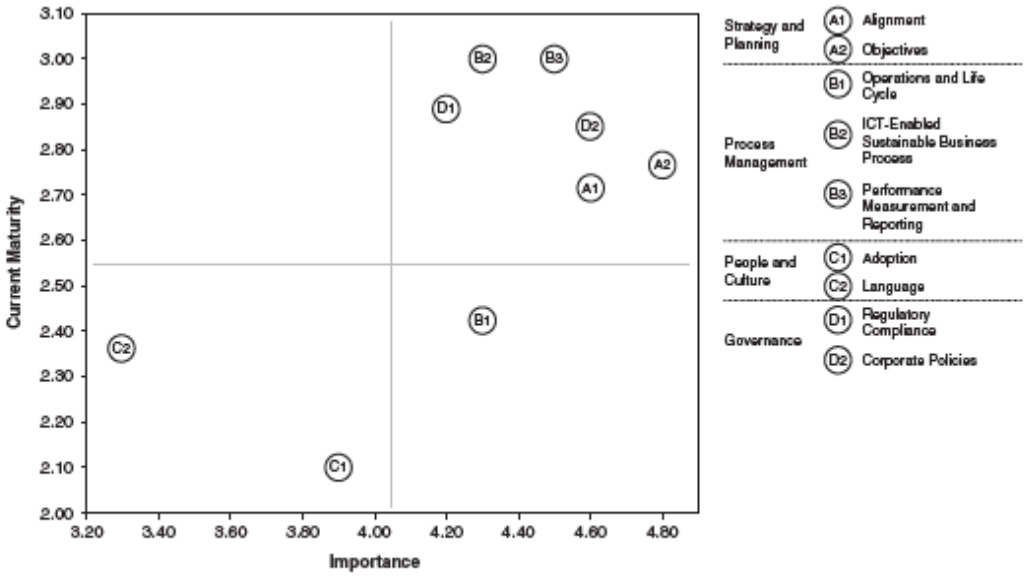


Figure 9.3 Maturity level versus importance plotted for each capability building block.

Figure below shows the consolidated survey results, resulting in an overall maturity level for each capability building block. This organization is close to level-3 maturity overall but is less mature in some individual capabilities. It views alignment and objectives under the strategy and planning category as the most important capability building blocks, but it has not achieved level-3 maturity in these areas. It also views operations and life cycle as important capabilities, but its maturity level for that building block is even lower (level 2).

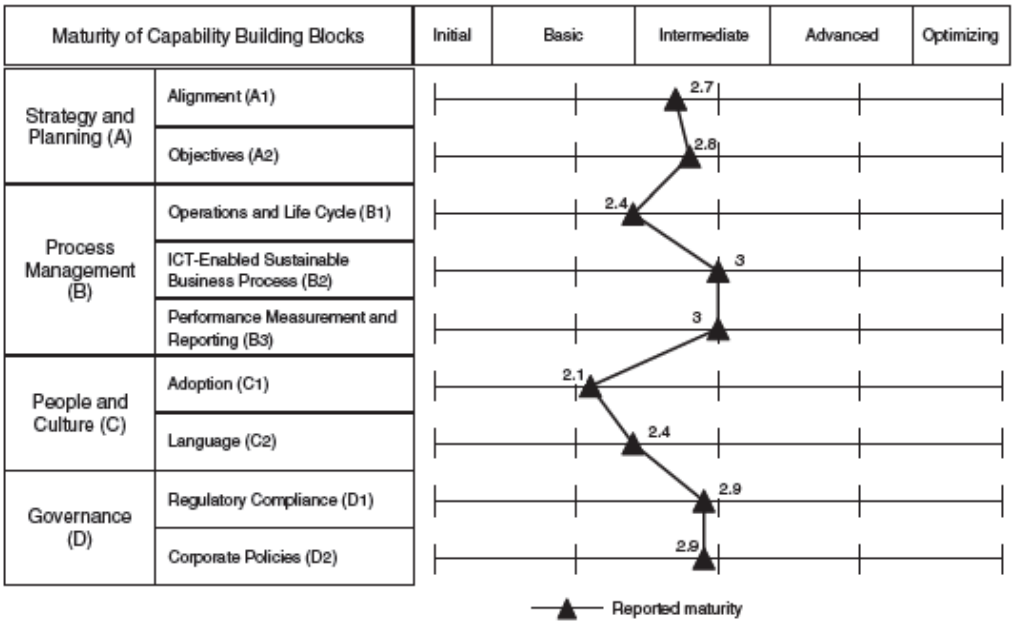


Figure 9.4 Aggregated results for the current maturity level from the assessment.

Assessing and Managing SICT Progress

With the initial assessment complete, organizations will have a clear view of current capability and key areas for action and improvement. However, to further develop SICT capability, the organization should assess and manage SICT progress over time by using the assessment results to achieve the following goals:

- Develop a roadmap and action plan.
- Add a yearly follow-up assessment to the overall IT management process to measure over time both progress and the value delivered from adopting SICT.

Agreeing on stakeholder ownership for each priority area is critical to developing both short-term and long-term action plans for improvement. The assessment results can be used to prioritize the opportunities for quick wins – that is, those capabilities that have smaller gaps between current and desired maturity and those that are recognized as more important but might have a bigger gap to bridge.